

◆ What is a Tag Helper in ASP.NET Core MVC?

📌 Definition

A **Tag Helper** in ASP.NET Core MVC is a **server-side component** that helps you generate HTML elements in Razor views **using C# logic** while keeping the syntax clean and HTML-like.

👉 It **extends standard HTML tags** by adding server-side behavior.

◆ Why Tag Helpers?

Before Tag Helpers, we used **HTML Helpers** like:

```
@Html.TextBoxFor(m => m.Name)
```

This is:

- Harder to read
- Not HTML-friendly

👉 Tag Helpers improve this:

```
<input asp-for="Name" />
```

- ✓ Looks like HTML
 - ✓ Easier for frontend + backend developers
 - ✓ Strongly typed (compile-time safety)
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◆ How Tag Helpers Work (Internally)

- Razor engine parses HTML
- Detects attributes like asp-for, asp-action
- Converts them into proper HTML during rendering

👉 Example:

```
<input asp-for="Name" />
```

→ Converts to:

```
<input type="text" id="Name" name="Name" value="" />
```

◆ Common Tag Helpers in ASP.NET Core MVC

Now let's go through **most used Tag Helpers** (important for interviews + real projects)

1 Anchor Tag Helper

Used to generate links based on routing.

```
<a asp-controller="Home" asp-action="Index">Home</a>
```

👉 Output:

```
<a href="/Home/Index">Home</a>
```

Attributes:

- asp-controller
 - asp-action
 - asp-route-id
-

2 Form Tag Helper

Used to create forms that post to controller actions.

```
<form asp-controller="Account" asp-action="Login" method="post">
```

👉 Automatically generates correct action URL

3 Input Tag Helper

Used for model binding inputs.

```
<input asp-for="Email" />
```

👉 Based on model:

```
public string Email { get; set; }
```

✓ Automatically sets:

- type
 - name
 - id
 - value
-

Label Tag Helper

```
<label asp-for="Email"></label>
```

👉 Output:

```
<label for="Email">Email</label>
```

5 **Validation Tag Helpers**

a) Validation Message

```
<span asp-validation-for="Email"></span>
```

b) Validation Summary

```
<div asp-validation-summary="All"></div>
```

👉 Works with model validation

6 **Select Tag Helper (Dropdown)**

```
<select asp-for="Country" asp-items="Model.CountryList"></select>
```

👉 Binds dropdown with model data

7 **Textarea Tag Helper**

```
<textarea asp-for="Description"></textarea>
```

9 Partial Tag Helper

```
<partial name="_Header" />
```

👉 Loads partial views

10 Cache Tag Helper

```
<cache expires-after="TimeSpan.FromSeconds(60)">
```

```
<p>Cached Content</p>
```

```
</cache>
```

👉 Improves performance

◆ Tag Helper vs HTML Helper

Feature	Tag Helper	HTML Helper
Syntax	HTML-like	C# method
Readability	High	Medium
IntelliSense	Better	Limited
Frontend Friendly	Yes	No
Compile-time Safety	Yes	Yes

◆ Types of Tag Helpers

1. Built-in Tag Helpers

Provided by ASP.NET Core:

- Anchor
- Form
- Input
- Label

- Validation
-

2. Custom Tag Helpers

You can create your own:

```
public class EmailTagHelper : TagHelper
{
    public string Address { get; set; }

    public override void Process(TagHelperContext context, TagHelperOutput output)
    {
        output.TagName = "a";
        output.Attributes.SetAttribute("href", "mailto:" + Address);
    }
}
```

Usage:

```
<email address="test@gmail.com"></email>
```

Key Advantages

- ✓ Clean HTML-like syntax
 - ✓ Strong model binding
 - ✓ Reusable components
 - ✓ Better separation of concerns
 - ✓ Easier maintenance
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